

	Answer	Mark
2(a)(i)	The only correct answer is A. <i>B is incorrect because TMV does not have an envelope</i> <i>C is incorrect because λ phage does not have an envelope</i> <i>D is incorrect because TMV does not have an envelope</i>	(1)

Question number	Answer	Mark
2(a)(ii)	The only correct answer is C. <i>A is incorrect because Ebola has a helical structure</i> <i>B is incorrect because HIV has a polyhedral structure</i> <i>D is incorrect because TMV has a helical structure</i>	(1)

Question number	Answer	Mark
2(a)(iii)	The only correct answer is C. <i>A is incorrect because Ebola, HIV and TMV all have RNA and λ phage has DNA</i> <i>B is incorrect because Ebola, HIV and TMV all have RNA and λ phage has DNA</i> <i>D is incorrect because Ebola, HIV and TMV all have RNA and λ phage has DNA</i>	(1)

Question number	Answer	Additional guidance	Mark
2(b)(i)	U G G U U C C G C (1)		(1)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An explanation that includes two of the following points: • (because the positive strand) has the {codons / codes} for the {proteins / amino acids} (1) • because the positive strand has the <u>complementary</u> base sequence needed to make the negative strand (1)	ACCEPT used in translation / (viral) genes negative strand does not have the correct codons DO NOT ACCEPT transcription IGNORE refs to sense and antisense (strands)	(2)

Question number	Answer	Additional guidance	Mark
2(c)	An explanation that includes four of the following points: • because (during 18 days) new viruses are produced (1) • (new) viruses {burst out of / damage} (host) cells (1) • infecting more cells / causing the spread of the virus (1) • takes time for the immune system to be stimulated (1) • so {person may become ill / virus out of control} before immune system stimulated (1)	IGNORE refs to latency / description of retroviruses / replication of DNA IGNORE names of host cells ACCEPT description of event that take place ACCEPT description e.g. not enough antibodies present for opsonisation ACCEPT before {receiving antiviral drugs / drugs can take effect} DO NOT ACCEPT kill virus	(4)